SETH-based Gaussian kernel density estimation hardness across dimension and scale regimes

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Editorial scope. The counted result is original Theorem-29 (Hardness of Gaussian KDE). All hardness conclusions are conditional on SETH as stated; supporting statements and alternate parameter specializations are not separately counted.

Allowed standard prerequisites (editorial).

- SETH-based cited closest-pair hardness, standard Schur polynomial identities and Cauchy-Binet. Author exact cited hardness statements and complete bibliography retained; all counting-matrix and kernel-specific local proofs included.

Definition 1.1 ((Approximate) Kernel Density Estimation $\mathsf{KDE}_f(n, m, \varepsilon, B)$)**.** *Let $f : [0, B] \to [0, 1]$ be a real function and define the (kernel) function $k(x, y) = f(\|x - y\|_2^2)$.*

Suppose we are given as input $2n$ points $x^{(1)}, \cdots, x^{(n)}, y^{(1)}, \cdots, y^{(n)} \in \mathbb{R}^m$ with the guarantee that $\|x^{(i)} - y^{(j)}\|_2^2 \leq B$ for all $i, j \in [n]$. Define the kernel matrix $K \in [0, 1]^{n \times n}$ by $K[i, j] = f(\|x^{(i)} - y^{(j)}\|_2^2)$ for $i, j \in [n]$. Then given additionally a vector $u \in \mathbb{R}^n$, one needs to output a vector $v \in \mathbb{R}^n$ such that $\|v - Ku\|_\infty \leq \varepsilon \|u\|_1$.

2 Preliminaries

2.1 SETH and known hard problems

We now introduce several variants of the Bichromatic Closest Pair problem.

Problem 2.1 (Hamming (Exact) Bichromatic Closest Pair)**.** $\mathsf{Hamming\text{-}BCP}(n, m)$*: Given two sets $A = \{x^{(1)}, \cdots, x^{(n)}\}, B = \{y^{(1)}, \cdots, y^{(n)}\} \subset \{0, 1\}^m$, compute $\min_{i,j \in [n]} \|x^{(i)} - y^{(j)}\|_2^2$.*

Theorem 2.2 ([AW15])**.** *Assuming SETH, for every $q \in (0, 1)$, there exists $C > 0$ such that if $m > C \log n$, then* $\mathsf{Hamming\text{-}BCP}(n, m)$ *cannot be solved in time $O(n^{2-q})$ for any constant $q > 0$.*

Similarly, one can define the Hamming approximate BCP problem and its decision version.

Problem 2.3 (Hamming Approximate BCP)**.** Hamming-Apx-BCP(n, m, μ)*: Given two sets* A, B *as in Problem 2.1 as well as* $\mu \in \mathbb{R}_+$*, output* $d \in \mathbb{R}$ *such that* $\min_{i,j\in[n]} \|x^{(i)} - y^{(j)}\|_2^2 \le d \le (1+\mu)\min_{i,j\in[n]} \|x^{(i)} - y^{(j)}\|_2^2$.

Theorem 2.4 ([Rub18])**.** *Assuming SETH, for every* $q > 0$*, there exist* $C > 0, \mu > 0$ *such that if* $m > C \log n$*, then* Hamming-Apx-BCP(n, m, μ) *cannot be solved in time* $O(n^{2-q})$ *for any constant* $q > 0$.

In the low dimensional regime $m = o(\log n)$, we consider the hardness of the ℓ_2 BCP problem.

Problem 2.5 (ℓ_2 (Exact) Bichromatic Closest Pair)**.** ℓ_2-BCP(n, m, D)*: Given two sets* $A = \{x^{(1)}, \cdots, x^{(n)}\}, B = \{y^{(1)}, \cdots, y^{(n)}\} \in \mathbb{Z}^m$ *such that* $\max_{i,j\in[n]} \|x^{(i)} - y^{(j)}\|_2^2 \le D$*, compute* $\min_{i,j\in[n]} \|x^{(i)} - y^{(j)}\|_2^2$.

Theorem 2.6 ([Che20])**.** *Assuming SETH, for every* $q > 0$*, there exists* $C_1, C_2 > 0$ *such that if* $m > C_1^{\log^* n}$ *and* $D > m^{C_2^{\log^* n} \cdot (\log n)/m}$*, then* ℓ_2-BCP(n, m, D) *cannot be solved in time* $O(n^{2-q})$ *for any constant* $q > 0$.

To unify the hardness results for BCP in different dimension regimes, we view the Hamming BCP problem as an ℓ_2 BCP problem with $D = m$. Formally, we combine Theorem 2.4 and Theorem 2.6 as follows.

Theorem 2.7. *Assuming SETH, for every* $q > 0$*, there exists* $C > 0$ *such that* ℓ_2-BCP(n, m, D) *cannot be solved in time* $O(n^{2-q})$ *for any constant* $q > 0$ *if either of the following holds: (1)* $m > C \log n, D = m$*, or (2)* $m > C^{\log^* n}, D > m^{C^{\log^* n} \cdot (\log n)/m}$.

2.2 Kernels of interest

In this work we focus primarily on three kernels $k(x, y) = f(\|x - y\|_2^2)$:

- Gaussian kernel $f(x) = e^{-x}$;
- Rational Quadratic kernel $f(x) = 1/(1+x)^\sigma$ for $\sigma \ge 1$ a parameter;
- t-Student kernel $f(x) = 1/(1+x^\rho)$ for $\rho \ge 1$ a parameter.[1]

Rational Quadratic kernel and t-Student kernel are two typical kernels with mild decay (as opposed to the rapid decay of Gaussian kernel). This property is abstracted by Backurs et al. [BCIS18] in the definition of a smooth kernel. Here we only focus on decreasing radial kernels.

Definition 2.8 (smooth kernel)**.** *A decreasing radial kernel* $k(x, y) = f(\|x - y\|_2^2)$ *is* (L, t) *smooth if for any* $0 < a < b$*,*

$$\frac{f(a)}{f(b)} \le L \left(\frac{b}{a}\right)^t.$$

By calculation one can verify that Rational Quadratic kernel and t-Student kernel are respectively $(1, 1)$- and $(1, \rho)$-smooth.

[1] t in the name of the kernel in principle should be the name of the parameter. We here use ρ as parameter while keeping the name *t-Student kernel* unchanged.

Positive definite kernels Most commonly-studied kernels are positive definite. We will find that our lower bound approach takes a particularly nice form for such kernels.

Definition 2.9 (Positive definite kernel)**.** *A kernel $k : \mathbb{R}^m \times \mathbb{R}^m \to \mathbb{R}$ is positive definite if for any n points $x_1, \cdots, x_n \in \mathbb{R}^m$, the Gram matrix $G = [k(x_i, x_j)]_{i,j\in[n]}$ is always positive definite.*

For radial kernels, we have the following concise characterization of positive definite kernels.

Definition 2.10. *Let $f : \mathbb{R}_{\geq 0} \to \mathbb{R}$ be a real function. We say f is absolutely monotone if $G \in C^\infty(\mathbb{R}_{\geq 0})$ and $f^{(k)}(t)$ for all $k \in \mathbb{N}$ and $t \geq 0$, and we say f is completely monotone if $G \in C^\infty(\mathbb{R}_{\geq 0})$ and $(-1)^k \cdot f^{(k)}(t) \geq 0$ for all $k \in \mathbb{N}$ and $t \geq 0$.*

Theorem 2.11 (Schoenberg's characterization)**.** *Let $f : \mathbb{R}_{\geq 0} \to \mathbb{R}$ be a real function. Then the kernel $k(x, y) = f(\|x - y\|_2^2)$ is positive definite if and only if f is completely monotone on $\mathbb{R}_{\geq 0}$.*

2.3 Tools from linear algebra

As in prior work, we will make use of the Cauchy-Binet formula.

Lemma 2.12 (Cauchy-Binet formula)**.** *Let $k > 0$ be an integer, and for functions $A : [k] \times \mathbb{N} \to \mathbb{R}$ and $B : \mathbb{N} \times [k] \to \mathbb{R}$, define the matrix $C \in \mathbb{R}^{k\times k}$ by, for $i, j \in [k]$, $C[i, j] = \sum_{\ell=0}^{\infty} A[i, \ell]B[\ell, j]$. If the sum converges absolutely for all i, j, then*

$$\det(C) = \sum_{1\leq \ell_1 < \cdots < \ell_k} \det(A[\ell_1, \cdots, \ell_k]) \cdot \det(B[\ell_1, \cdots, \ell_k]).$$

Here $A[\ell_1, \cdots, \ell_k]$(resp. $B[\ell_1, \cdots, \ell_k]$) is the $k\times k$ matrix obtained from A (resp. B) by taking the columns (resp. rows) $\ell_1, \cdots, \ell_k$.

3 Main reduction from BCP

Most of our new lower bounds are based on the reduction below from Hamming or ℓ_2 Exact Bichromatic Closest Pair to KDE. This reduction generalizes a framework developed in [ACSS20] to accommodate different parameter regimes. In this section we will give the outline of the reduction and establish an upper bound on $1/\varepsilon$ determined by the quantity $\tau(M)$.

Definition 3.1. *Let $D > 0$ be an integer. Fix vectors $\alpha, \beta \in \mathbb{R}^D$ and $c \in \mathbb{R}$. Suppose $c + \alpha_\ell\beta_p \in [0, 1]$ for all $\ell, p \in [D]$. Then for function $f : [0, 1] \to \mathbb{R}$, the counting matrix $M = M(D; f, \alpha, \beta)$ is a $D \times D$ matrix defined by $M[\ell, p] = f(c + \alpha_\ell\beta_p), \ell, p \in [D]$.*

Theorem 3.2. *Let $\alpha \in \mathbb{R}^D$ be a fixed vector and $\beta \in \mathbb{R}^D$ be the identity vector defined by $\beta_p = p$. If $\mathsf{KDE}_f(n, m, \varepsilon)$ can be solved in $T(n, m, \varepsilon)$ time, then ℓ_2-$\mathsf{BCP}(n, m, D)$ with $m = n^{o(1)}, D = n^{o(1)}$ can be solved in $T(n, m +$*

$1, (3n\tau(M))^{-1}) \cdot n^{o(1)} + n^{1+o(1)}$ *time, where* M *is the* $D \times D$ *counting matrix associated with* f, α, β *and*

$$\tau(M) = \max_{0 \neq b \in \mathbb{R}^D} \frac{\|M^{-1}b\|_\infty}{\|b\|_\infty}.$$

Proof. Given two sets $X = \{x^{(1)}, \cdots, x^n\}, Y = \{y^{(1)}, \cdots, y^{(n)}\} \subseteq \mathbb{Z}^m$ such that $\max_{i,j\in[n]} \|x^{(i)} - y^{(j)}\|_2^2 \leq D$, let $W \in \mathbb{N}^{D\times n}$ denote the *distance count matrix* defined by $W[p, i] = \#[j \in [n] : \|x^{(i)} - y^{(j)}\|_2^2 = p]$. Then the matrix product $U = M \times W$ gives

$$\begin{aligned} U[\ell, i] &= \sum_{p=1}^{D} M[\ell, p] \cdot W[p, i] \\ &= \sum_{p=1}^{D} f\big(c + \alpha_\ell \beta_p\big) \sum_{j\in[n]} \mathbb{1}\left[\|x^{(i)} - y^{(j)}\|_2^2 = p\right] \\ &= \sum_{j\in[n]} f\left(c + \alpha_\ell \cdot \beta\left[\|x^{(i)} - y^{(j)}\|_2^2\right]\right). \end{aligned} \tag{3.1}$$

We note that when β is the identity vector, the summation (3.1) can be formulated as a KDE instance. More specifically, let $\tilde{x}_\ell^{(i)}, \tilde{y}_\ell^{(j)} \in \mathbb{R}^{m+1}$ be defined by $\tilde{x}_\ell^{(i)}[k] = \sqrt{\alpha(\ell)}x^{(i)}[k]$ if $k \in [m]$ and $\tilde{x}_\ell^{(i)}[m+1] = \sqrt{c}$; $\tilde{y}_\ell^{(j)}[k] = \sqrt{\alpha(\ell)}y^{(j)}[k]$ if $k \in [m]$ and $\tilde{y}_\ell^{(j)}[m+1] = 0$. Then

$$U[\ell, i] = \sum_{j\in[n]} f(\|\tilde{x}_\ell^{(i)} - \tilde{y}_\ell^{(j)}\|_2^2).$$

Therefore we have the following algorithm for ℓ_2-$\mathsf{BCP}(n, m, D)$ using KDE_f as a subroutine. On input $X = \{x^{(1)}, \cdots, x^{(n)}\}, Y = \{y^{(1)}, \cdots, y^{(n)}\}$ such that $\max_{i,j\in[n]} \|x^{(i)} - y^{(j)}\|_2^2 \leq D$:

1. For $\ell \in [D]$, construct vectors $\tilde{x}_\ell^{(i)}, \tilde{y}_\ell^{(j)}, i, j \in [n]$. Then approximate the ℓ-th row of U: $\hat{U}[\ell] \approx U[\ell] = K_\ell \times \mathbf{1}$ using the KDE_f oracle.
2. Compute $\hat{W} = M^{-1} \times \hat{U}$, and round each entry to the closest integer.

We claim that if we call the KDE_f subroutine with $\varepsilon = (3n\tau(M))^{-1}$, then the distance count matrix W is exactly recovered after the rounding step. Indeed, for fixed $i \in [n]$, letting $W[\cdot, i], U[\cdot, i]$ respectively denote the i-th column of matrix W and U, we have

$$\|\hat{W}[\cdot, i] - W[\cdot, i]\|_\infty = \|M^{-1}(\hat{U}[\cdot, i] - U[\cdot, i])\|_\infty \leq \tau(M) \cdot \|\hat{U}[\cdot, i] - U[\cdot, i]\|_\infty.$$

If the KDE_f subroutine guarantees that $\|\hat{U}[\ell] - U[\ell]\|_\infty \leq (3n\tau(M))^{-1} \cdot \|\mathbf{1}\|_1 = (3\tau(M))^{-1}$ for all $\ell \in \{1, \cdots, D\}$, then the entry-wise difference between $\hat{W}$ and W is bounded by $1/3$.

The D calls to the subroutine then take in total $D \cdot T(n, m, (3n\tau(M))^{-1})$ time. It takes in addition $D \cdot O(nm) = n^{1+o(1)}$ operations to construct the vectors and $O(D^\omega)$ time for step 2.[2] □

[2]We always assume $f(x)$ can be exactly computed in constant time for any $x \in [0, 1]$.

We now combine the reduction above and the hardness result of BCP in Theorem 2.7. The hardness of the KDE problem follows.

Theorem 3.3. *Assuming SETH, for every $q > 0$, there exists $C \geq 0$ such that $\mathsf{KDE}_f(n, m, (3n\tau(M))^{-1})$ cannot be solved in time $O(n^{2-q})$ for any constant $q > 0$. if either of the following holds: (1) $m > C \log n, D = m$ or (2) $m > C^{\log^* n}, D > m^{C^{\log^* n} \cdot (\log n)/m}$. Here M is the $D \times D$ counting matrix in Definition 3.1.*

4 Schur polynomials

Let $\mathbb{F}$ be a field. For matrices $A, B \in \mathbb{F}^{m \times n}$, let $A \circ B$ denote the Hadamard product of A and B, i.e. the $m \times n$ matrix with entries $(A \circ B)[i, j] = A[i, j] \cdot B[i, j], i \in [m], j \in [n]$. For matrix $A \in \mathbb{F}^{m \times n}$ we define the Hadamard powers $A^{\circ 1} = A$ and $A^{\circ(k+1)} = A^{\circ k} \circ A$ for integer $k \geq 1$. Similarly one can define the Hadamard product and Hadamard power for vectors $u \in \mathbb{F}^m$. Moreover, given a vector $u \in \mathbb{F}^m$ and a tuple $r = (r_1, \cdots, r_n) \in \mathbb{N}^n$, we denote by $u^{\circ r}$ the $m \times n$ matrix

$$\begin{pmatrix} | & | & & | \\ u^{\circ r_1} & u^{\circ r_2} & \cdots & u^{\circ r_n} \\ | & | & & | \end{pmatrix} = \begin{pmatrix} u_1^{r_1} & u_1^{r_2} & \cdots & u_1^{r_n} \\ \vdots & \vdots & \ddots & \vdots \\ u_m^{r_1} & u_m^{r_2} & \cdots & u_m^{r_n} \end{pmatrix}.$$

In particular, when $m = n$, we call $u^{\circ r}$ a generalized Vandermonde matrix. This is a natural generalization of the (usual) Vandermonde matrix $u^{\circ \delta}$ associated with the tuple $\delta = (0, 1, \cdots, m-1)$.

The concept of Schur polynomials was first proposed by Cauchy and defined as the ratio of two generalized Vandermonde determinants. In what follows we denote by $\mathbb{N}^m_< = \{x \in \mathbb{N}^m : x_1 < \cdots < x_m\}$ the set of m-tuples composed of distinct entries in ascending order, and define $\mathbb{N}^m_\leq = \{x \in \mathbb{N}^m : x_1 \leq \cdots \leq x_m\}$ similarly.

Definition 4.1 (Cauchy's definition of Schur polynomials)**.** *Let $m > 0$ be an integer and $\lambda = (\lambda_1, \cdots, \lambda_m) \in \mathbb{N}^m_\leq$ an integer tuple. We define the Schur polynomial on variables $(u_1, \cdots, u_m)$ by*

$$s_\lambda(u) = s_\lambda(u_1, \cdots, u_m) = \frac{\det(u^{\circ(\lambda+\delta)})}{\det(u^{\circ\delta})} = \frac{\det(u^{\circ(\lambda+\delta)})}{V(u)}.$$

Here $V(u) = \det(u^{\circ\delta}) = \prod_{1 \leq i < j \leq m}(u_j - u_i)$ is the Vandermonde determinant.

It is known that Schur polynomial has an equivalent definition by Littlewood using Young tableaux.

Definition 4.2. *Let $m > 0$ be an integer, and $\lambda = (\lambda_1, \cdots, \lambda_m) \in \mathbb{N}^m_\leq$ an integer tuple. A semi-standard Young tableaux (SSYT) of shape λ on alphabet $[m]$ is a left-aligned two-dimensional rectangular array T of cells, with λ_i cells in the i-th row ($i \in [m]$), from bottom to top, such that*

- *each cell in T is assigned with an entry from $1, \cdots, m$;*
- *entries weakly decrease in each row, from left to right;*

- *entries strictly decrease in each column, from top to bottom.*

Moreover, for a SSYT T, we define the type $\mathrm{t}(T) = (t_1, \cdots, t_m) \in \mathbb{N}^m$ *of* T, *where* t_j *is the number of cells in* T *assigned with entry* $j \in [m]$.

Definition 4.3 (Littlewood's definition of Schur polynomials)**.** *Let* $m > 0$ *be an integer. For integer tuple* $\lambda = (\lambda_1, \cdots, \lambda_m) \in \mathbb{N}^m_{\leq}$, *the Schur polynomial* $s_\lambda(u)$ *can be equivalently defined by*

$$s_\lambda(u) = \sum_{T \in \mathcal{T}} u^{\mathrm{t}(T)},$$

where $\mathcal{T}$ *is the set of all SSYT of shape* λ *on alphabet* $[m]$.

5 Direct calculation of $\tau(M)$: Gaussian kernel and t-Student kernel

5.1 Gaussian kernel

In this section we show upper bounds for

$$\tau(M) = \max_{b \in \mathbb{R}^D : b \neq 0} \frac{\|M^{-1}b\|_\infty}{\|b\|_\infty} \leq D \cdot \max_{t,s \in [D]} |M^{-1}[t,s]|$$

where M is the counting matrix associated with the Gaussian kernel $f(x) = e^{-Bx}$. By standard facts in linear algebra, we can rewrite

$$|M^{-1}[t,s]| = \left| \frac{\det(M^{s-}_{t-})}{\det(M)} \right|$$

with M^{s-}_{t-} the submatrix of M consisting of all entries but those in row s or column t.

For the simplicity of analysis, we study the counting matrix M of dimension $(D+1)$. A slight modification of the main reduction can employ such a matrix to recover the distance matrix with a redundant row: $W[p,i] = \#[\|x^{(i)} - y^{(j)}\|_2^2 = p], i \in [n], p = \{0, 1, \cdots, D\}$.

Theorem 5.1. *Let* $\alpha \in \mathbb{R}^{D+1}$ *be a fixed vector and* $\beta \in \mathbb{R}^{D+1}$ *be the identity vector. Let* M *be the* $(D+1) \times (D+1)$ *counting matrix associated with Gaussian kernel* $f = e^{-Bx}$ *and* α, β. *Then there exists a vector* α *such that*

$$\tau(M) \leq \left(\frac{5e}{1 - e^{-B/D}} \right)^D.$$

Proof. Let $x_i = \exp(-B\alpha_i)$, then $M = [x_i^j : i, j \in \{0, \cdots, D\}]$ is a Vandermonde matrix with

$$|\det(M)| = |\det(x^{\circ(0,\cdots,D)})| = V(x).$$

On the other hand, we observe that M^{s-}_{t-} can be viewed as a generalized Vandermonde matrix. Making use of both the algebraic and combinatorial definition

of Schur polynomials, we have[3]

$$|\det(M_{t-}^{s-})| = |\det((x^{s-})^{\circ(0,\cdots,t-1,t+1,\cdots,D)})| = V(x^{s-})\left(\sum_{R\subseteq\{0,\cdots,D\}\setminus\{s\}}\prod_{r\in R}x^r\right),$$

in which x^{s-} denotes the vector $(x_0,\cdots,x_{s-1},x_{s+1},\cdots,x_D)$ and

$$\sum_{R\subseteq\{0,\cdots,D\}\setminus\{s\}}\prod_{r\in R}x^r \le \prod_{i\in\{0,\cdots,D\}\setminus\{s\}}(1+x_i) \le 2^D.$$

Therefore

$$\max_{s,t}|\det(M_{t-}^{s-})| \le 2^D\cdot\max_s V(x^{s-}) = 2^D\cdot V(x)\left(\min_s\prod_{i\in[D]\setminus\{s\}}|x_i-x_s|\right)^{-1}.$$

We now pick $0=\alpha_0<\cdots<\alpha_{D-1}<\alpha_D=1/D$ such that $|x_{i+1}-x_i| = |\exp(-B\alpha_{i+1})-\exp(-B\alpha_i)| = \frac{1}{D}(1-e^{-B/D})$. Let $r=\frac{1}{D}(1-e^{-B/D})$, then

$$\begin{aligned}\prod_{i\in\{0,\cdots,D\}\setminus\{s\}}|x_i-x_s| &= \left(\prod_{i=0}^{s-1}(s-i)r\right)\left(\prod_{i=s+1}^{D}(i-s)r\right)\\ &= r^D\cdot s!(D-s)! \ge r^D\cdot(\frac{D}{2e})^{D/2\cdot 2} = (\frac{1-e^{-B/D}}{2e})^D.\end{aligned}$$

Combining the calculations together, we have

$$\tau(M) \le D\cdot\max_{s,t}\left|\frac{\det(M_{t-}^{s-})}{\det(M)}\right| \le D\cdot 2^D\left(\min_s\prod_{i\in[D]\setminus\{s\}}|x_i-x_s|\right)^{-1} \le \left(\frac{5e}{1-e^{-B/D}}\right)^D.$$

□

The following hardness result for Gaussian KDE therefore arises from a combination of the general KDE hardness Theorem 3.3 and the bound on $\tau(M)$ above.

Theorem 5.2 (Hardness of Gaussian KDE)**.** *Let $f(x)=e^{-Bx}$ be the Gaussian kernel with $B\ge 1$. Then assuming SETH, for every $q>0$, there exists $C_1,C_1',C_2,C_3,C_3',C_4\ge 0$ such that $\mathsf{KDE}_f(n,m,\varepsilon)$ cannot be solved in time $O(n^{2-q})$ if either of the following holds:*

(1) $m>C_1\log n, B<C_1'\log n, 1/\varepsilon>(C_2m/B)^m$, or

(2) $C_3\log^* n<m<C_3'(\log n)/(\log\log n)$, $\log\log(1/\varepsilon)>(\log n)/m\cdot(\log m)\cdot C_4^{\log^* n}$.

Proof. For regime (1), first pick appropriate C_1' so that $B/m<1$.

$$3n\tau(M) = 3n\left(\frac{5e}{1-e^{-B/m}}\right)^m \le 3n\left(\frac{5e}{B/(em)}\right)^m = 3\cdot 2^{C_1^{-1}m}\cdot\left(\frac{5e^2m}{B}\right)^m \le \left(\frac{C_2m}{B}\right)^m$$

[3]The Schur polynomial here in fact equals an elementary symmetric polynomial.

for some constant $C_2 > 0$. For regime (2), we have $B/D \leq 1$ and thus

$$3n\left(\frac{5e}{1-e^{-B/D}}\right)^D \leq 3n\left(\frac{5e^2D}{B}\right)^D \leq 3n(5e^2D)^D.$$

If $m < (\log n)/(\log\log n)$,

$$\log D = (\log n)/m \cdot (\log m) \cdot C^{\log^* m} = (\log\log n)^2(1-o(1)) \cdot C^{\log^* m} > \log\log n.$$

$$\log(3n\tau(M)) \leq \log(3n) + D\log(5e^2D) < C'D\log(5e^2D)$$

For some $C' > 0$. Thus,

$$\log\log(3n\tau(M)) \leq \log D + \log\log(5e^2D) \leq (\log n)/m \cdot (\log m) \cdot C_4^{\log^* m}.$$

□

5.2 t-Student kernel

For t-Student kernels $f(x) = 1/(1+x^\rho)$, we prove a similar bound on $\tau(M)$. The key observation here is that both M and M^{t-}_{s-} are (scaled) Cauchy matrices. For vectors $a, b \in \mathbb{R}^n$, the Cauchy matrix associated with a, b is defined by $M[i,j] = 1/(a_i + b_j), i, j \in [n]$. The following closed-form formula is known for its determinant.

Theorem 5.3 (Cauchy determinant)**.** *Let $n \geq 1$ be an integer. For $a, b \in \mathbb{R}^n$, let $M = [1/(a_i + b_j)]_{i,j\in[n]}$ be the associated Cauchy matrix. Then*

$$\det(M) = \frac{\prod_{1\leq i<j\leq n}(a_i - a_j)(b_i - b_j)}{\prod_{i,j\in[n]}(a_i + b_j)}.$$

Corollary 5.4. *For $a, b \in \mathbb{R}^n$ with $a_i, b_j \in (0,1), \forall i, j \in [n]$, and $s, t \in [n]$, we have*

$$\begin{aligned}\det\left[\frac{1}{1+a_ib_j}\right]_{i,j\in[n]} &= \left(\prod_{i=1}^n \frac{1}{a_i}\right)\det\left[\frac{1}{a_i^{-1}+b_j}\right] \\ &= \left(\prod_{i=1}^n \frac{1}{a_i}\right)\frac{\prod_{1\leq i<j\leq n}(a_i^{-1}-a_j^{-1})(b_i-b_j)}{\prod_{i,j\in[n]}(a_i^{-1}+b_j)} = \frac{\prod_{1\leq i<j\leq n}(a_j-a_i)(b_i-b_j)}{\prod_{i,j\in[n]}(1+a_ib_j)}.\end{aligned}$$

$$\left|\frac{\det[(1+a_ib_j)^{-1}]_{i,j\in[n]}}{\det[(1+a_ib_j)^{-1}]_{i\in[n]^{s-},j\in[n]^{t-}}}\right| = \left|\frac{\prod_{i\in[n]^{s-}}(a_i-a_s)\prod_{j\in[n]^{t-}}(b_j-b_t)}{\prod_{i\in[n]}(1+a_ib_t)\prod_{j\in[n]^{t-}}(1+a_sb_j)}\right|.$$

Here $[n]^{s-} = [n]\backslash\{s\}, [n]^{t-} = [n]\backslash\{t\}$.

Before proving the bound on $\tau(M)$, we gather several simple inequalities to be used.

Lemma 5.5. *(I) Let $r \geq 1, a, b > 0$ be real numbers. Then $(a+b)^r \geq a^r + b^r$. (II) Let $a, b \geq 0$ be integers. Then $a!b! \geq ((\lfloor\frac{a+b}{2}\rfloor)!)^2$.*

Proof. (I) Taking the derivative, one can show $f(x) = (1+x)^r - x^r$ is increasing in x for $x > 0$. Thus $f(b/a) = (1 + b/a)^r - (b/a)^r \geq f(0) = 1$. The inequality follows.

(II) For $a = b = 0$, the inequality is trivial. If not, we have $\binom{a+b}{a} \leq \binom{a+b}{\lceil (a+b)/2 \rceil}$, $a!b! \geq (\lceil \frac{a+b}{2} \rceil)!(\lfloor \frac{a+b}{2} \rfloor)! \geq ((\lfloor \frac{a+b}{2} \rfloor)!)^2$. □

Theorem 5.6. *Let $\rho \geq 1$ be a real number. Let $\alpha \in \mathbb{R}^D$ be a fixed vector and $\beta \in \mathbb{R}^D$ be the identity vector. Let M be the $D \times D$ counting matrix associated with t-Student kernel $f(x) = 1/(1 + x^\rho)$ and α, β. Then there exists a vector α such that*

$$\tau(M) \leq (7e)^{2\rho D}.$$

Proof. The counting matrix is

$$M = [f(\alpha_i \beta_j)]_{i,j \in [D]} = \left[\frac{1}{1 + (\alpha_i \beta_j)^\rho} \right]_{i,j \in [D]}$$

For simplicity we assume D is odd. Let $s, t \in [D]$. By corollary 5.4, we have

$$\left| \frac{\det(M)}{\det(M^{s-}_{t-})} \right| = \left| \frac{\prod_{i \in [D]^{s-}} (\alpha_i^\rho - \alpha_s^\rho) \prod_{j \in [D]^{t-}} (\beta_j^\rho - \beta_t^\rho)}{\prod_{i \in [D]} (1 + \alpha_i^\rho \beta_t^\rho) \prod_{j \in [D]^{t-}} (1 + \alpha_s^\rho \beta_j^\rho)} \right|.$$

We set $\alpha = \beta$ to be the scaled identity vector with $\alpha_i = \beta_i = i/D, \forall i \in [D]$. (By rescaling $\alpha_i = i/D^2, \beta_i = i$ one can make β the identity vector.) Then

$$\prod_{i \in [D]} (1 + \alpha_i^\rho \beta_t^\rho) \prod_{j \in [D]^{t-}} (1 + \alpha_s^\rho \beta_j^\rho) \leq 2^{2D-1},$$

$$\begin{aligned} \prod_{i \in [D]^{s-}} |\alpha_i^\rho - \alpha_s^\rho| &= \prod_{i=1}^{s-1} \frac{s^\rho - i^\rho}{D^\rho} \prod_{i=s+1}^{D} \frac{i^\rho - s^\rho}{D^\rho} \geq \prod_{i=1}^{s-1} \frac{(s-i)^\rho}{D^\rho} \prod_{i=s+1}^{D} \frac{(i-s)^\rho}{D^\rho} \\ &= \left(\frac{(s-1)!(D-s)!}{D^{D-1}} \right)^\rho \geq \left(\frac{(\lfloor \frac{D-1}{2} \rfloor)!}{D^{D-1}} \right)^\rho \geq \left(\frac{D-1}{2eD} \right)^{\rho(D-1)} \geq (3e)^{-\rho D}. \end{aligned}$$

Similarly,

$$\prod_{j \in [D]^{t-}} |\beta_j^\rho - \beta_t^\rho| \geq (3e)^{-\rho D}.$$

In conclusion, we have

$$\tau(M) \leq D \cdot \max_{s,t} \left| \frac{\det(M^{s-}_{t-})}{\det(M)} \right| \leq D \cdot 2^{2D} \cdot (3e)^{2\rho D} \leq (7e)^{2\rho D}.$$

□

Combining the bound on $\tau(M)$ above and the general KDE hardness Theorem 3.3, we obtain

Theorem 5.7 (Hardness results of t-Student KDE)**.** *Let $f(x) = 1/(1 + x^\rho)$ be a t-Student kernel parameterized by $\rho \geq 1$ an absolute constant. Then assuming SETH, for every $q > 0$, there exists $C_1, C_2, C_3, C_3', C_4 \geq 0$ such that $\mathsf{KDE}_f(n, m, \varepsilon)$ cannot be solved in time $O(n^{2-q})$ if either of the following holds:*

(1) $m > C_1 \log n, 1/\varepsilon > n^{C_2}$, *or*

(2) $C_3 \log^* n < m < C_3'(\log n)/(\log\log n)$, $\log\log(1/\varepsilon) > (\log n)/m \cdot (\log m) \cdot C_4^{\log^* n}$.

Here C_1, C_3, C_3' *are absolute constants while* C_2, C_4 *are dependent on* ρ.

Proof. For regime (1),

$$3n\tau(M) = 3n(7e)^{2\rho m} = 3n(7e)^{2\rho \cdot C_1 \log n} = n^{C_2}$$

for some C_2 dependent on ρ. For regime (2), if $m < (\log n)/(\log\log n)$,

$$\log D = (\log n)/m \cdot (\log m) \cdot C^{\log^* m} = (\log\log n)^2(1 - o(1)) \cdot C^{\log^* m} > \log\log n.$$

$$\log(3n\tau(M)) = \log(3n) + 2\log(7e)\rho D < C_\rho' D,$$

$$\log\log(3n\tau(M)) < (\log n)/m \cdot (\log m) \cdot C_4^{\log^* m}.$$

for some C_4 dependent on ρ. □

6 Cauchy-Binet Expansion

To bound $\tau(M)$ for counting matrices that are associated with more general kernels, we establish the connection between the determinant $\det(M)$ and the Taylor coefficients of f via Cauchy-Binet Expansion, based on a framework given in [ACSS20]. Our improvement mainly derives from an observation that the terms in the expansion are in fact generalized Vandermonde determinants. The additional structure in the determinants allows for simplifications in analysis through Schur polynomials.

Define infinite matrices $A : [D] \times \mathbb{N} \to \mathbb{R}$, $\tilde{A} : [D] \times \mathbb{N} \to \mathbb{R}$, and $B : \mathbb{N} \times [D] \to \mathbb{R}$ by

$$A[i,k] = \alpha_i^k, \quad \tilde{A}[i,k] = \frac{f^{(k)}(c)}{k!}\alpha_i^k, \quad B[k,j] = \beta_j^k.$$

Suppose the Taylor series of $f(c+x)$ at $c \in [0,1]$ is *absolutely convergent* whenever $c + x \in [0,1]$. Then for $i, j \in [D]$, assuming $c + \alpha_i\beta_j \in [0,1]$, we have

$$M[i,j] = f(c + \alpha_i\beta_j) = \sum_{k=0}^{\infty} \frac{f^{(k)}(c)}{k!}\alpha_i^k\beta_j^k = \sum_{k=0}^{\infty} \tilde{A}[i,k]B[k,j].$$

By Cauchy-Binet formula, we expand $\det(M)$ as follows.

$$\begin{aligned}
\det(M) &= \sum_{\bar{n}\in\mathbb{N}^D_<} \det \tilde{A}[[D];\bar{n}]\cdot\det B[\bar{n};[D]] \\
&= \sum_{\bar{n}\in\mathbb{N}^D_<} \left(\prod_{n\in\bar{n}}\frac{f^{(n)}(c)}{n!}\right)\det A[[D];\bar{n}]\cdot\det B[\bar{n};[D]] \\
&= \sum_{\bar{n}\in\mathbb{N}^D_<} \left(\prod_{n\in\bar{n}}\frac{f^{(n)}(c)}{n!}\right)\det(\alpha^{\circ\bar{n}})\det(\beta^{\circ\bar{n}}) \\
&= \sum_{\lambda\in\mathbb{N}^D_\leq} \left(\prod_{i=1}^{D}\frac{f^{(\lambda_i+\delta_i)}(c)}{(\lambda_i+\delta_i)!}\right)\det(\alpha^{\circ(\lambda+\delta)})\det(\beta^{\circ(\lambda+\delta)}) \quad (\lambda=\bar{n}-\delta) \\
&= V(\alpha)V(\beta)\sum_{\lambda\in\mathbb{N}^D_\leq} \left(\prod_{i=1}^{D}\frac{f^{(\lambda_i+\delta_i)}(c)}{(\lambda_i+\delta_i)!}\right)s_\lambda(\alpha)s_\lambda(\beta). \qquad (6.1)
\end{aligned}$$

The last step follows from the algebraic definition of Schur polynomials. Similarly,

$$\det(M^{s-}_{t-}) = V(\alpha^{s-})V(\beta^{t-})\sum_{\lambda\in\mathbb{N}^{D-1}_\leq} \left(\prod_{i=1}^{D-1}\frac{f^{(\lambda_i+\delta_i)}(c)}{(\lambda_i+\delta_i)!}\right)s_\lambda(\alpha^{s-})s_\lambda(\beta^{t-}). \quad (6.2)$$

6.1 Absolutely monotonic kernels

At this point one may try bounding (6.1) and (6.2) using analytic properties of Schur polynomials. However, for a general function f, the arbitrary signs of Taylor coefficients largely complicate the analysis, making tight bounds out of reach. Hence we first study the easy case where $\tau(M)$ is associated with an *absolutely monotonic* kernel f, i.e., $f:[0,1]\to[0,1]$ satisfies $f^{(n)}(c)\geq 0$ for all $n\in\mathbb{N}$ and $c>0$. We will shortly see that such kernels are in fact not artificial as all the positive definite kernels naturally reduce to this case.

We first use the following indentity of Schur polynomials to "align" (6.1) and (6.2).

Proposition 6.1. *If $\lambda_1\neq 0$, then $s_{(\lambda_1,\cdots,\lambda_n)}(\alpha_1=0,\alpha_2,\cdots,\alpha_n)=0$. If $\lambda_1=0$, then $s_{(\lambda_1,\cdots,\lambda_n)}(\alpha_1=0,\alpha_2,\cdots,\alpha_n)=s_{(\lambda_2,\cdots,\lambda_n)}(\alpha_2,\cdots,\alpha_n)$.*

Proof. By the combinatorial definition of Schur polynomials, we have

$$s_{(\lambda_1,\cdots,\lambda_n)}(\alpha_1,\alpha_2,\cdots,\alpha_n)=\sum_{T\in\mathcal{T}}\alpha^{\mathrm{t}(T)}$$

where $\mathcal{T}$ is the set of all SSYT of shape $\lambda=(\lambda_1,\cdots,\lambda_n)$ on alphabet $[n]$, and $\mathrm{t}(T)$ denotes the type of SSYT T. Now set $\alpha_1=0$. If the letter 1 appears in a SSYT T, i.e. $\mathrm{t}(T)[1]>0$, then the corresponding monomial vanishes when evaluated. Therefore the equation simplifies to

$$s_{(\lambda_1,\cdots,\lambda_n)}(\alpha_1,\alpha_2,\cdots,\alpha_n)=\sum_{T\in\mathcal{T}_1}\alpha^{\mathrm{t}(T)}$$

where $\mathcal{T}_1$ is the set of all SSYT of shape λ on alphabet $\{2, 3, \cdots, n\}$.

However, if $\lambda_1 > 0$, no sequence of length n on alphabet $\{2, 3, \cdots, n\}$ satisfies the (strictly) decreasing constraint in the first column of the tableau. In this case $s_{(\lambda_1,\cdots,\lambda_n)}(\alpha_1, \alpha_2, \cdots, \alpha_n) = 0$. If $\lambda_1 = 0$, then $\mathcal{T}_1$ is essentially the set of all SSYT of shape $\lambda' = (\lambda_2, \cdots, \lambda_n)$ over alphabet $\{2, 3, \cdots, n\}$. By definition,

$$s_{(\lambda_1,\cdots,\lambda_n)}(\alpha_1, \alpha_2, \cdots, \alpha_n) = \sum_{T\in\mathcal{T}_1} \alpha^{\mathrm{t}(T)} = s_{(\lambda_2,\cdots,\lambda_n)}(\alpha_2, \cdots, \alpha_n).$$

□

In consequence, if we fix $\alpha = (\alpha_1, \alpha_2, \cdots, \alpha_D)$ with $\alpha_1 = 0$, then

$$\begin{aligned}
\det(M) &= V(\alpha)V(\beta) \sum_{\lambda\in\mathbb{N}^D_{\le}} \left(\prod_{i=1}^{D} \frac{f^{(\lambda_i+\delta_i)}(c)}{(\lambda_i+\delta_i)!}\right) s_\lambda(\alpha)s_\lambda(\beta) \\
&= V(\alpha)V(\beta) \sum_{\lambda^{1-}\in\mathbb{N}^{D-1}_{\le}} \left(f(c)\prod_{i=2}^{D} \frac{f^{(\lambda_i+\delta_i)}(c)}{(\lambda_i+\delta_i)!}\right) s_{\lambda^{1-}}(\alpha^{1-})s_{\lambda^{1-}}(\beta^{1-}) \\
&= V(\alpha)V(\beta)f(c) \sum_{\lambda\in\mathbb{N}^{D-1}_{\le}} \left(\prod_{i=1}^{D-1} \frac{f^{(\lambda_i+\delta_{i+1})}(c)}{(\lambda_i+\delta_{i+1})!}\right) s_\lambda(\alpha^{1-})s_\lambda(\beta^{1-}).
\end{aligned}$$

Here $\lambda^{1-}, \alpha^{1-}, \beta^{1-}$ are obtained by removing the first entry in the corresponding partition/vector λ, α, β. (In the last step we abuse the notation by renaming λ^{1-} as λ.) Meanwhile,

$$\begin{aligned}
\max_{s,t} \det(M^{s-}_{t-}) &\le \max_{s,t} V(\alpha^{s-})V(\beta^{t-}) \cdot \max_{s,t} \sum_{\lambda\in\mathbb{N}^{D-1}_{\le}} \left(\prod_{i=1}^{D-1} \frac{f^{(\lambda_i+\delta_i)}(c)}{(\lambda_i+\delta_i)!}\right) s_\lambda(\alpha^{s-})s_\lambda(\beta^{t-}) \\
&= \left(\max_{s,t} V(\alpha^{s-})V(\beta^{t-})\right) \sum_{\lambda\in\mathbb{N}^{D-1}_{\le}} \left(\prod_{i=1}^{D-1} \frac{f^{(\lambda_i+\delta_i)}(c)}{(\lambda_i+\delta_i)!}\right) s_\lambda(\alpha^{1-})s_\lambda(\beta^{1-}).
\end{aligned}$$

In what follows, we let

$$E_\lambda = \left(\prod_{i=1}^{D-1} \frac{f^{(\lambda_i+\delta_{i+1})}(c)}{(\lambda_i+\delta_{i+1})!}\right) s_\lambda(\alpha^{1-})s_\lambda(\beta^{1-}),\ \ F_\lambda = \left(\prod_{i=1}^{D-1} \frac{f^{(\lambda_i+\delta_i)}(c)}{(\lambda_i+\delta_i)!}\right) s_\lambda(\alpha^{1-})s_\lambda(\beta^{1-})$$

denote the corresponding terms in the two sums. For absolutely monotonic function f, we have $F_\lambda > 0$ and $E_\lambda > 0$ for all $\lambda \in N^{D-1}_{\le}$. Thus,

$$\max_{s,t} \frac{\det(M^{s-}_{t-})}{\det(M)} \le \frac{1}{f(c)} \max_{s,t} \frac{V(\alpha^{s-})V(\beta^{t-})}{V(\alpha)V(\beta)} \max_{\lambda\in\mathbb{N}^{D-1}_{\le}} \frac{F_\lambda}{E_\lambda}. \tag{6.3}$$

6.2 KDE hardness for positive definite kernels

To accommodate positive definite kernels, we slightly modify the reduction in Section 3. Let $k(x, y) = f(\|x - y\|_2^2)$ be a positive definite kernel. By Schoenberg's characterization (Theorem 2.11), we have $(-1)^k \cdot f^{(k)}(t) \ge 0$ for all

$k \in \mathbb{N}, t \geq 0$. Then for the function $g(x) = f(1-x)$, it holds $g^{(k)}(t) = (-1)^k \cdot f^{(k)}(1-t) \geq 0$ for all $k \in \mathbb{N}, t \in [0,1]$. Namely g is an absolutely monotonic kernel.

Next we see how KDE_f can be related to $\tau(M_g)$. Let M be the $D \times D$ counting matrix associated with g, and let $W \in \mathbb{N}^{D \times n}$ be a re-indexed distance count matrix defined by $W[p,i] = \#[j \in [n] : \|x^{(i)} - y^{(j)}\|_2^2 = D - p]$. Then the matrix product $U = M \times W$ gives

$$\begin{aligned} U[\ell, i] &= \sum_{p=1}^{D} M[\ell, p] \cdot W[p, i] \\ &= \sum_{p=1}^{D} g(\alpha_\ell \beta_p) \sum_{j \in [n]} \mathbb{1}\left[\|x^{(i)} - y^{(j)}\|_2^2 = D - p\right] \\ &= \sum_{j \in [n]} g\left(c + \alpha_\ell \cdot \beta\left[D - \|x^{(i)} - y^{(j)}\|_2^2\right]\right). \end{aligned} \tag{6.4}$$

If β is the identity vector, then

$$\begin{aligned} U[\ell, i] &= \sum_{j \in [n]} g\left(c + \alpha_\ell \cdot (D - \|x^{(i)} - y^{(j)}\|_2^2)\right) \\ &= \sum_{j \in [n]} f\left(1 - c - \alpha_\ell D + \alpha_\ell \|x^{(i)} - y^{(j)}\|_2^2\right). \end{aligned}$$

Constructing vectors $\tilde{x}_\ell^{(i)}, \tilde{y}_\ell^{(j)} \in \mathbb{R}^{m+1}$ defined by $\tilde{x}_\ell^{(i)}[k] = \sqrt{\alpha(\ell)} x^{(i)}[k]$ if $k \in [m]$ and $\tilde{x}_\ell^{(i)}[m+1] = \sqrt{1 - c - \alpha_\ell D}$; $\tilde{y}_\ell^{(j)}[k] = \sqrt{\alpha(\ell)} y^{(j)}[k]$ if $k \in [m]$ and $\tilde{y}_\ell^{(j)}[m+1] = 0$, we have

$$U[\ell, i] = \sum_{j \in [n]} f(\|\tilde{x}_\ell^{(i)} - \tilde{y}_\ell^{(j)}\|_2^2).$$

In this way we show that the new product U can also be computed using KDE subroutines, and the reduction proceeds the same way as in Section 3 to recover the (re-indexed) distance count matrix. By similar analysis, we have the following hardness result relating the complexity of KDE_f to $\tau(M_g)$.

Theorem 6.2. *Let $f : [0,1] \to [0,1]$ be a function and $g(x) = f(1-x)$. Then assuming SETH, for every $q > 0$, there exists $C \geq 0$ such that $\mathsf{KDE}_f(n, m, (3n\tau(M))^{-1})$ cannot be solved in time $O(n^{2-q})$ for any constant $q > 0$. if either of the following holds: (1) $m > C \log n, D = m$ or (2) $m > C^{\log^* n}, D > m^{C^{\log^* n} \cdot (\log n)/m}$. Here M is the $D \times D$ counting matrix associated with function g.*

6.3 Hardness of Rational Quadratic kernel

For f, α, β of certain forms, we give explicit bounds on the two terms in (6.3). Regarding the ratio of Vandermonde determinants, we focus on $V(x^{s-})/V(x)$ for scaled identity x defined by $x_p = p/D$.

Proposition 6.3. *Let $\rho \geq 1$ be a real number and $x \in [0,1]^D$ be the vector defined by $x_i = i/D$. Then for $s \in [D]$ we have*

$$\frac{V(x^{s-})}{V(x)} = \prod_{i \in [D] \setminus \{s\}} |x_i - x_s|^{-1} \leq (3e)^D.$$

Proof.

$$\prod_{i\in[D]\setminus\{s\}} |x_i - x_s| = \prod_{i=1}^{s-1} \frac{s-i}{D} \cdot \prod_{i=s+1}^{D} \frac{i-s}{D}$$
$$= \frac{(s-1)!(D-s)!}{D^{D-1}} \geq \frac{((\frac{D-1}{2})!)^2}{D^{D-1}} \geq \left(\frac{D-1}{2eD}\right)^{(D-1)} \geq (3e)^{-D}.$$

□

Hence for vectors α defined by $\alpha_\ell = \ell/D, \ell \in [D]$ and β defined by $\beta_p = p/D, p \in [D]$, we have

$$\max_{s,t} \frac{V(\alpha^{s-})V(\beta^{t-})}{V(\alpha)V(\beta)} \leq (3e)^{2D}.$$

For the term involving Taylor coefficients, we fix a real number $\sigma \geq 1$ and focus on the absolutely monotonic function $f(x) = (2-x)^{-\sigma}$. By calculation, for $k \in \mathbb{N}$, $f^{(k)}(x) = (2-x)^{-(\sigma+k)} \prod_{i=0}^{k-1}(\sigma+i)$. Then for $c = 0$, we have

$$\frac{F_\lambda}{E_\lambda} = \left(\prod_{i=1}^{D-1} \frac{f^{(\lambda_i+\delta_i)}(0)}{(\lambda_i+\delta_i)!}\right) \Big/ \left(\prod_{i=1}^{D-1} \frac{f^{(\lambda_i+\delta_{i+1})}(0)}{(\lambda_i+\delta_{i+1})!}\right) = \prod_{i=1}^{D-1} \left(2 \cdot \frac{1+\lambda_i+\delta_i}{\sigma+\lambda_i+\delta_i}\right) \leq 2^{D-1}.$$

Combining the calculations, we obtain the following bound on $\tau(M)$.

Theorem 6.4. *Let $\sigma \geq 1$ be a real number. Let $f : [0,1] \to [0,1]$ be the function $f(x) = (2-x)^{-\sigma}$, and let vectors $\alpha, \beta \in \mathbb{R}^D$ be defined by $\alpha_\ell = \ell/D, \beta_p = p/D$. Then the $D \times D$ counting matrix M associated with f, α, β has*

$$\tau(M) \leq D \cdot 2^\sigma \cdot (3e)^{2D} \cdot 2^{D-1} \leq 2^\sigma (7e)^{2D}.$$

Combining the bound on $\tau(M)$ associated with $f(x) = (1+(1-x))^{-\sigma}$ with the hardness of positive definite KDE Theorem 6.2, we obtain

Theorem 6.5 (Hardness of Rational Quadratic KDE)**.** *Let $f(x) = 1/(1+x)^\sigma$ be a Rational Quadratic kernel parameterized by $\sigma \geq 1$ an absolute constant. Then assuming SETH, for every $q > 0$, there exists $C_1, C_2, C_3, C_3', C_4 \geq 0$ such that $\mathsf{KDE}_f(n, m, \varepsilon)$ cannot be solved in time $O(n^{2-q})$ for any constant $q > 0$. if either of the following holds:*

(1) $m > C_1 \log n, 1/\varepsilon > n^{C_2}$, or

(2) $C_3 \log^* n < m < C_3'(\log n)/(\log\log n), \log\log(1/\varepsilon) > (\log n)/m \cdot (\log m) \cdot C_4^{\log^* n}$.

Here C_1, C_3, C_3' are absolute constants while C_2, C_4 are dependent on σ.

Proof. For regime (1),

$$3n\tau(M) = 3n \cdot 2^\sigma (7e)^{2m} = n^{C_2}$$

for some C_2 dependent on ρ. For regime (2), if $m < (\log n)/(\log\log n)$,

$$\log D = (\log n)/m \cdot (\log m) \cdot C^{\log^* m} > \log\log n.$$
$$\log(3n\tau(M)) = \log(3n) + \sigma + 2\log(7e)\rho D < C_\sigma' D,$$
$$\log\log(3n\tau(M)) < (\log n)/m \cdot (\log m) \cdot C_4^{\log^* m}.$$

for some C_4 dependent on ρ. □

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